KAIZEN MATHS

Lesson starter

Normal Distribution and Standardisation

Any Normal Distribution

Name _____

S1

X ~ N(150, 100). Find P(X < 155.2).

S2

X ~ N(150, 100). Find P(141.6 < X < 158.4).

S3

X ~ N(50, 36). Find a such that P(X < a) = 0.7500.

S4

X ~ N(80, 64). Find P(|X - 80| > 8).

Show your working

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STA-9E35-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Normal Distribution and Standardisation

Any Normal Distribution

Name

E1

X ~ N(60, 100). Find P(X < 76.4).

E2

X ~ N(200, 100). Find P(191.6 < X < 212.8).

E3

X ~ N(80, 64). Find a such that P(X < a) = 0.7500.

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Show your working

Teacher answers & working

Normal Distribution and Standardisation · Any Normal Distribution

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S1

X ~ N(150, 100). Find P(X < 155.2).

Answer:

0.6985

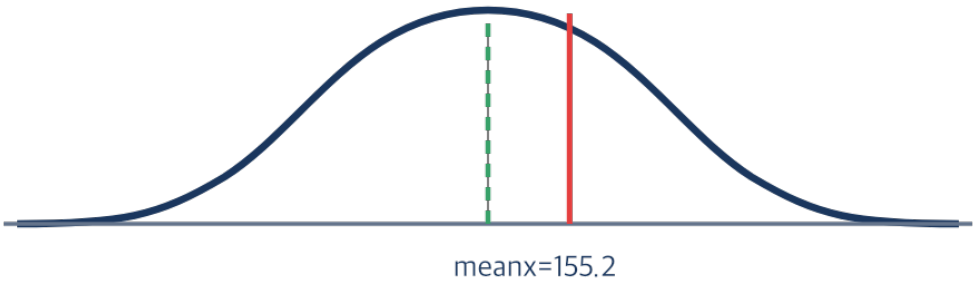
Working:

Identify the mean and standard deviation.

$\mu = 150$, $\sigma^2 = 100$, so $\sigma = 10$.

Standardise the boundary value.

$Z = (X - \mu) / \sigma = (155.2 - 150) / 10 = 0.52$.



$P(X < 155.2) = P(Z < 0.52)$.

$P(Z < 0.52) = \Phi(0.52)$.

Using tables/calculator gives 0.6985.

0.6985

Teacher answers & working

Normal Distribution and Standardisation · Any Normal Distribution

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S2

X ~ N(150, 100). Find P(141.6 < X < 158.4).

Answer:

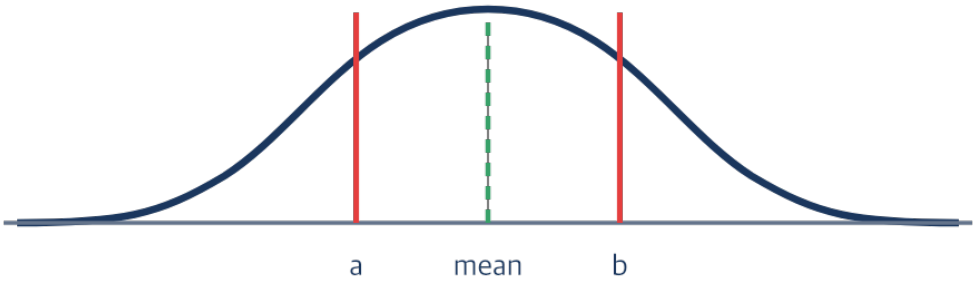
0.5991

Working:

Standardise both boundary values.

$z_1 = (141.6 - 150) / 10 = -0.84$.

$z_2 = (158.4 - 150) / 10 = 0.84$.



$P(141.6 < X < 158.4) = P(-0.84 < Z < 0.84)$.

$= \Phi(0.84) - \Phi(-0.84)$.

$= 0.7995 - 0.2005 = 0.5991$.

0.5991

Teacher answers & working

Normal Distribution and Standardisation · Any Normal Distribution

STA-9E35-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

X ~ N(50, 36). Find a such that P(X < a) = 0.7500.

Answer:

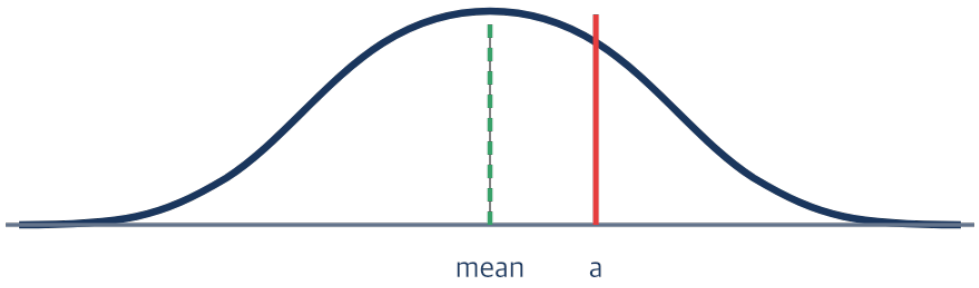
$a = 54$

Working:

Convert the left-tail probability into a z-value.

P(X < a) = 0.7500 means P(Z < z) = 0.7500.

$z = \Phi^{-1}(0.7500) \approx 0.67.$



Convert back from z to x.

$x = \mu + z\sigma.$

$a = 50 + (0.67)(6) = 54.$

$a = 54$

Teacher answers & working

Normal Distribution and Standardisation · Any Normal Distribution

STA-9E35-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

X ~ N(80, 64). Find P(|X - 80| > 8).

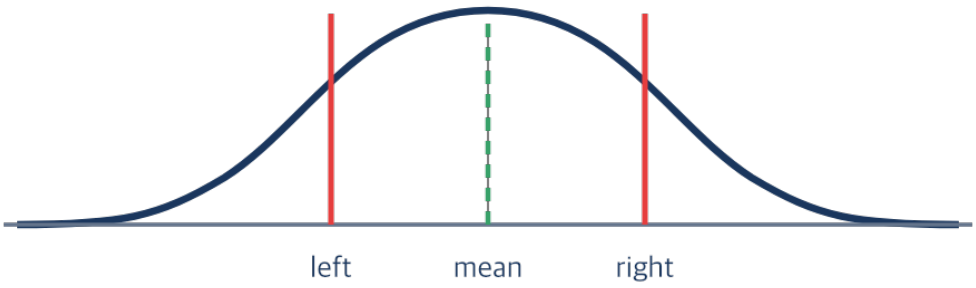
Answer:

0.3173

Working:

Split the absolute-value statement into two tails.

|X - 80| > 8 means X < 72 or X > 88.



Standardise both boundaries.

$z_1 = (72 - 80) / 8 = -1.00$, and $z_2 = (88 - 80) / 8 = 1.00$.

$P(|X - 80| > 8) = P(Z < -1.00) + P(Z > 1.00)$.

$= 2(1 - \Phi(1.00)) = 0.3173$.

0.3173

Teacher answers & working

Normal Distribution and Standardisation · Any Normal Distribution

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E1

X ~ N(60, 100). Find P(X < 76.4).

Answer:

0.9495

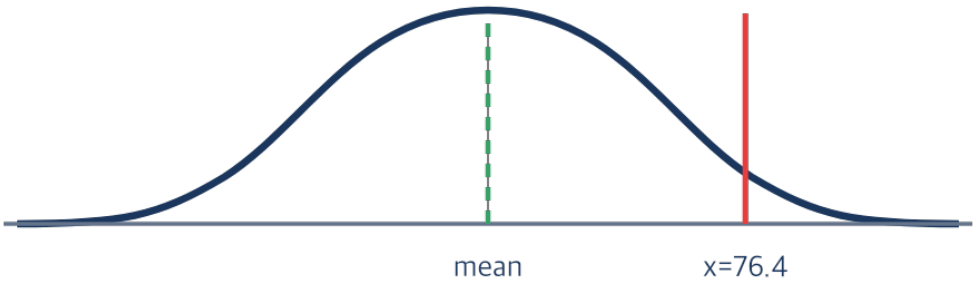
Working:

Identify the mean and standard deviation.

$\mu = 60$, $\sigma^2 = 100$, so $\sigma = 10$.

Standardise the boundary value.

$Z = (X - \mu) / \sigma = (76.4 - 60) / 10 = 1.64$.



$P(X < 76.4) = P(Z < 1.64)$.

$P(Z < 1.64) = \Phi(1.64)$.

Using tables/calculator gives 0.9495.

0.9495

Teacher answers & working

Normal Distribution and Standardisation · Any Normal Distribution

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E2

X ~ N(200, 100). Find P(191.6 < X < 212.8).

Answer:

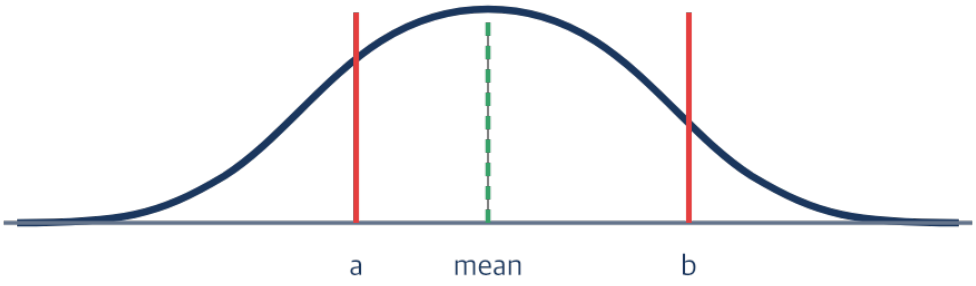
0.6993

Working:

Standardise both boundary values.

$z_1 = (191.6 - 200) / 10 = -0.84$.

$z_2 = (212.8 - 200) / 10 = 1.28$.



$P(191.6 < X < 212.8) = P(-0.84 < Z < 1.28)$.

$= \Phi(1.28) - \Phi(-0.84)$.

$= 0.8997 - 0.2005 = 0.6993$.

0.6993

Teacher answers & working

Normal Distribution and Standardisation · Any Normal Distribution

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E3

X ~ N(80, 64). Find a such that P(X < a) = 0.7500.

Answer:

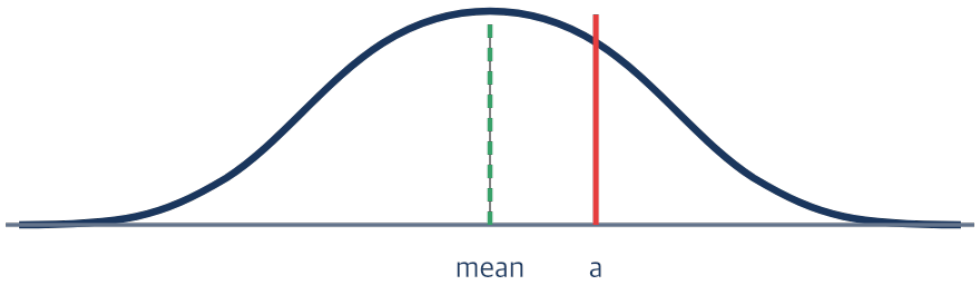
$a = 85.4$

Working:

Convert the left-tail probability into a z-value.

P(X < a) = 0.7500 means P(Z < z) = 0.7500.

$z = \Phi^{-1}(0.7500) \approx 0.67.$



Convert back from z to x.

$x = \mu + z\sigma.$

$a = 80 + (0.67)(8) = 85.4.$

$a = 85.4$